



Automated Box Folding Operation

CARDBOARD IN, BOX OUT: ENHANCING PACKAGING THROUGH AUTOMATION



Problem Statement

Current packaging operations require manual folding of flat cardboard into top-open boxes, a labor-intensive process that produces inconsistent folds, slows production throughput, increases ergonomic strain on workers, and limits the scalability of operations

Goal: design an automated packaging system that converts flat cardboard blanks into fully folded, top-open boxes ready for filling

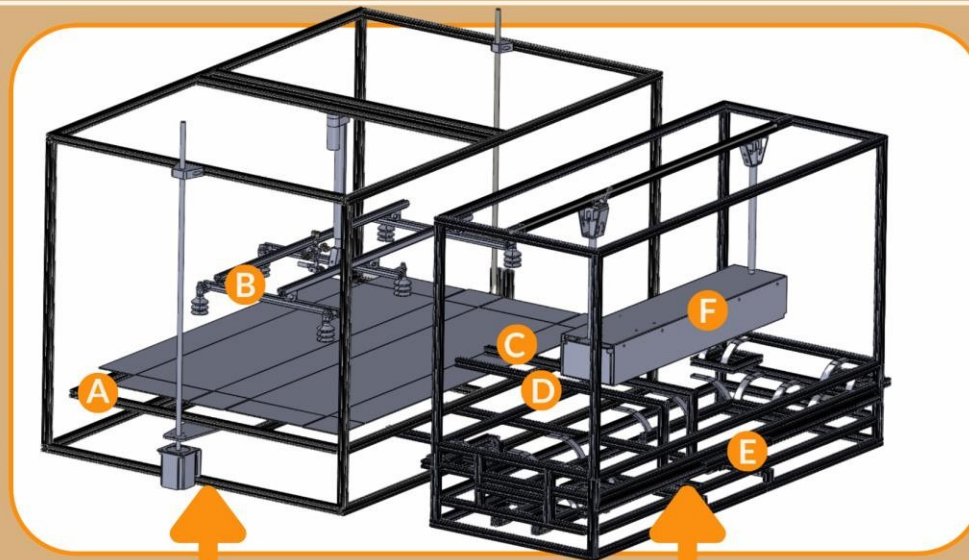
Background research on similar products

Design individual components of the system in SolidWorks

Combine components into 1 model

Build a physical prototype

Conduct testing



Lift System

- A. Lifts stack of cardboard up
- B. One cardboard blank picked up by vacuum motor
- C. Glue applied to flaps

Mold & Press

- D. Hook takes a single cardboard blank and positions it
- E. Adjustable mold where flat cardboard blank is positioned
- F. Press comes down onto the mold to form a box

Results

Projected Labor Savings
(after one year):

\$314,400

- Prototype Cycle Time: ~ 76 sec
- Projected Cycle Time: ~ 30 sec

Recommendations

New Technology for Hook

- Cost: ~ \$10,000

Incorporate Gluing System

- Cost: \$23-25k for prototype

Increase Stability

- Make out of steel
- Avoid 3D printed parts